The Lean Startup AI Assistant — Take-Home Booklet

> *Build your AI-Powered Digital COO. Learn to create your own commands and skills.*

How to Use This Booklet

Work through it front to back during the workshop, or use it as a reference after. Every exercise has a **prompt to run**, an **example output**, and space to paste your own. The 🔧 pages teach you to turn prompts into permanent tools.

Part 1 — The Framework

The AI-COO Concept

A solo founder is the CEO, the analyst, the marketer and the ops team at once, and the operational grind is what crowds out the actual product. An AI CLI — `claude` or `opencode` — acts as a tireless Chief Operating Officer that lives in your terminal, reads your project files, writes documents, analyzes data, and helps you decide at startup speed. Unlike a chat window, the CLI thinks in **workflows**: it works against your real files and its outputs are versionable.

The Lean Startup Loop + AI

```
┌──────────────┐
│   BUILD      │  hypotheses, interview scripts, landing pages
│  (hypothesize)│
└──────┬───────┘
       │
       ▼
┌──────────────┐
│   MEASURE    │  competitive analysis, survey analysis, insights
│  (get data)  │
└──────┬───────┘
       │
       ▼
┌──────────────┐
│   LEARN      │  pivot / persevere / zoom-in, sprint plan
│  (decide)    │
└──────┬───────┘
       │
       └────────►  back to BUILD (next iteration)
```

Setup (short version)

- **Claude Code:** `npm install -g @anthropic-ai/claude-code`, then `claude`
- **OpenCode:** `go install github.com/opencode-ai/opencode@latest`
- Set `ANTHROPIC_API_KEY` or log in via the browser prompt
- Verify: `bash setup/verify-setup.sh`

Your First Prompt — the 5 parts

> **Role** *("You are a Lean Startup strategist")* + **Context** *(your startup)* + **Task** *(what to produce)* + **Format** *(markdown table, numbered list)* + **Constraints** *(bootstrap budget, solo founder).*

🔧 Custom Commands (reference)

A command is a markdown file in `.claude/commands/`. Filename → command name. - `$ARGUMENTS` — replaced by whatever you type after the command - `!command` — injects live shell output (e.g. `!cat CLAUDE.md`) - Scope: `.claude/commands/` (project, shared via Git) vs. `~/.claude/commands/` (global, personal)

🔧 Skills (reference)

A skill is a folder `.claude/skills/<name>/` with a `SKILL.md` (YAML frontmatter + instructions). Claude **auto-activates** it when your task matches. Skills can bundle templates and examples.

	Commands	Skills
Trigger	Manual (`/x`)	Auto (context-detected)
Scope	One task	A whole domain
Form	One file	Folder + SKILL.md + files
Analogy	A tool you pick up	A skill in your brain

Part 2 — The Exercises

(Full prompts and example outputs live in `../exercises/` *and* `../sample-outputs/`*. Print those alongside this booklet, or read them in the repo.)*

Part 3 — The Case Study: Kinsight

> *Fictional founder, fictional company, synthetic data. Written for teaching only.*

Ahmad Saad — a solo founder, former healthcare data engineer, and parent of a 15-year-old — validates a platform connecting parents and child mental-health clinicians in one week using AI, and builds a reusable command library and skill as he goes.

Day	Produced	Built
Mon	Hypotheses, interview script, landing brief (15 min)	`/hypothesis`
Wed	Competitive + survey analysis, insights (20 min)	`analyze-survey`, `competitive-analysis`
Thu	Pivot decision, sprint plan (10 min)	`pivot-decision`, `sprint-plan`
Fri	SOPs, checklists, dashboard (10 min)	`generate-sop` + `lean-startup` skill

Total AI-assisted work: ~55 min. Estimated without AI: 2-4 weeks.

The Turn

Ahmad's bet was an **AI risk score** telling parents how their teen is doing — the instinct of someone who has spent a decade making systems measurable. His data disagreed:

- **76%** of 50 parents would not accept an algorithmic score about their own child
- Only **12%** rank a score as their top need; **64%** just want to know *what to say*
- **56%** would pay $5-10/month, only **14%** would pay $20+ — real but shallow
- **85%** of 20 clinicians name intake and documentation as their worst pain
- **70%** of clinicians would pay a per-seat licence; **0/20** get between-session signal
- **3 of 5** competitors already ship a score; **0 of 5** connect parent to clinician

Decision: ZOOM IN. Right thesis, wrong lever. Drop the score. Ship the clinician intake summary and the parent conversation guide — and move the payer to the clinic.

6 Lessons from Ahmad

1. Start with hypotheses, not features.
2. Let data decide — he killed his own favourite idea.
3. Zoom in fast (one afternoon, not one quarter).
4. Automate the ops grind.
5. Iterate weekly.
6. **Build tools, not just prompts** — his `.claude/` folder is a permanent asset.

The 7th Lesson, and the one this domain adds

The data didn't just say *wrong feature*. It said **don't build the diagnostic.** The riskiest thing to build was also the thing users didn't want and he couldn't responsibly validate. In health, those two tend to arrive together — listening to customers and behaving responsibly pointed the same way.

Which is why the last thing to learn is where AI **isn't** the tool. Use it for operational documents that are cheap to fix. Don't use it where being confidently wrong hurts someone and you can't tell that it's wrong — safeguarding procedures, clinical claims, fear-based copy aimed at frightened parents. Your AI will draft all three happily, and they will look completely reasonable.

Part 4 — The Prompt Playbook

25 reusable prompts across Strategy, Marketing, Finance, Operations, and Research. Full text in `../prompt-playbook/`. Swap the **[brackets]** or convert to commands with `$ARGUMENTS`.

Part 5 — Command & Skill Quick Reference

See `quick-reference-card.md` for the one-page cheat sheet.

Resources

- **Repo:** github.com/alhaol/ai-powered-digital-coo-research-assistant
- **Claude Code docs:** commands & skills
- **Recommended reading:** *The Lean Startup* (Ries), *The Mom Test* (Fitzpatrick), *Zero to One* (Thiel)

Dr. Ibrahim Abualhaol · AI Technical Lead · Ottawa, Canada · 18 July 2026